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Revision of
Int. Fed. Spec. L-T-004lb(GSA-FSS)
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Interim Revision of
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July 16, 1953

INTERIM FEDERAL SPECIFICATION

TABLE TOP AND FURNITURE ASSEMBLIES

(PLASTIC THERMOSETTING RESIN LAMINATE)

This Interim Federal Specification was developed by the Federal Supply Service, General Services Administration, Washington 25, D. C., based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized Federal agencies to use this Interim Federal Specification as a valid exception to Federal Specification L-T-4la.

1. SCOPE AND CLASSIFICATION

1.1 Scope.-This specification covers four types of table top and furniture assemblies. Types I and II are constructions of a decorative plastic laminate bonded to a plywood core, with a backing sheet of essentially the same material bonded to the other side of the plywood. The edges of types I and II are generally covered with a metal or plastic trim. Types III and IV are constructions of a decorative laminated top surface and plain laminated bottom surface molded or bonded directly to a wood particle core. The edges of Type III and IV are shaped or covered with a metal or plastic trim.

1.2 Classification.

1.2.1 Types.-Table top and furniture assemblies covered by this specification shall be of the following types, as specified:

Type I - Laminates bonded to a plywood core with resorcinol, phenolic, melamine or melamine-urea resins.

Type II - Laminates bonded to a plywood core with urea resin or any combination of urea and melamine resins.

Type III - Laminates molded to a fine grain wood particle core with phenolic resin. Core density is 53 to 63 pounds per cubic foot.

Type IV - Laminates bonded to a wood particle board core with urea or phenolic resin adhesive. Core density is 38 to 48 pounds per cubic foot.

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1.2.2 Trim.

1.2.2.1 Types I, II, and IV.-The composition and color of the metal or plastic trim shall be as specified.

1.2.2.2 Types III and IV.- When shaped the edge profile and color of the finish shall be as specified.

1.2.3 Sizes (length and width).-The sizes of the table top and furniture assemblies shall be as specified.

1.2.4 Thickness.-The thickness of the table top and furniture assemblies shall be as specified.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATION

2.1 Specifications and Standards.-The following Federal Specifications and Standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

L-P-406	-	Plastics, Organic; General Specifications, Test Methods.
L-P-508	-	Plastic Sheet, Laminated. (Thermosetting Resin).
PPP-B-591	-	Boxes, Fiberboard, Wood-Cleated.
PPP-B-636	-	Boxes, Fiber.
PPP-B-585	-	Boxes; Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes; Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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2.2 Other publications.-The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Commercial Standards:

CS35-56 - Hardwood Plywood.
CS45-55 - Douglas Fir Plywood.

(Copies of above publication may be obtained upon application to the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

Uniform Freight Classification Rules.

(Application for copies of the publication referenced above should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Decorative laminate.

3.1.1 Types I, II, and IV.-The decorative laminate for Types I, II, and IV assemblies shall conform to the requirements of Style D of Federal Specification L-P-508. Type, grade, class, finish, color, pattern, and nominal thickness of the decorative laminate shall be as specified in the contract or order.

3.1.2 Type III.-The decorative laminate for Type III assembly shall consist of a multilayer of thermosetting resin impregnated fiber to be molded integrally with a core and a balanced backing material. The laminate after molding shall meet the surface requirements of the decorative laminate, Style D, Federal Specification L-P-508, as applicable to type, grade, class and finish.

3.2 Plywood.-Plywood shall be hardwood with face and back veneers as specified in Grade 1 of Commercial Standard CS 35-56 or Douglas fir or spruce with face and back veneers of close grained wood as specified in Grade A of Commercial Standard CS 45-55. Plywood shall be a minimum of 5 plies and shall be sanded to the thickness as specified in the contract or order.

3.2.1 Type I, waterproof bond.-Waterproof bond shall be fabricated with phenolic, resorcinol, melamine, or any combination of melamine and urea resins and delaminate not more than 25 percent in any joint when tested as specified in 4.3.3.1.

3.2.2 Type II, water-resistant bond.-Water-resistant bond shall be fabricated with urea resin or any combination of urea and melamine resins and delaminate not more than 25 percent in any joint when tested as specified in 4.3.3.2.

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3.2.3 Alternatively, when specified, the plywood for a Type II assembly shall be of the following construction: a middle plank made of ponderosa pine core blocks glued together, having a hardwood cross band (approximately 1/8 inch before sanding) glued to the top and another similar 1/8 inch hardwood cross band glued to the back of the plank. The nominal thickness of the plywood shall be as specified.

3.2.4 Alternatively, when specified, the plywood for a Type II assembly shall be of the following construction: Unless otherwise specified, plywood shall be 5 ply lumber core. Lumber core to consist of Basswood or Poplar strips not to exceed 3½ inches in width and edge glued under pressure with urea resin adhesive. Cross bands shall be 1/16 inch thick of either Basswood or Poplar unless otherwise specified. Crossbands shall be sound veneer of random widths with edges jointed and glued. Face and back veneers shall conform to paragraph 4.7.3 Sound Grade (2) of Commercial Standard CS 35-56. The nominal thickness of the plywood shall be as specified.

3.3 Backing sheet.

3.3.1 Types I, II and IV.-The backing sheet for Types I, II and IV assemblies shall conform to the requirements of backing sheet covered by Federal Specification L-P-508, Style ND, Type V. Thickness shall be as specified.

3.3.2 Type III.-The backing sheet for type III assembly shall be a multilayer of thermosetting resin impregnated fiber.

3.3.3 When specified, the backing sheet may be omitted for a Type IV assembly providing that the contractor furnishes a certificate stating that the assembly will not warp or twist.

3.4 Impact resistance.-Not more than one of the ten test points shall show a crack at the surface, when tested as specified in 4.3.2.

3.5 Resistance to wetting and drying.

3.5.1 Type I.-The assembly shall delaminate not more than 25 percent in any resin adhesive bond (between plywood and decorative laminate, between plywood and backing, or in the plywood) when tested as specified in 4.3.3.1.

3.5.2 Type II (with 5-ply plywood core).-The assembly shall delaminate not more than 25 percent in any resin adhesive bond (between plywood and decorative laminate, between plywood and backing, or in the plywood) when tested as specified in 4.3.3.2.

3.5.3 Type II (with 3-ply plywood core), Type III, and Type IV.-The assembly shall show no visible delamination between any two layers of veneer nor between the plastic laminate faces and the plywood or core panel, totaling more than 2 inches in continuous length and over 3/16 inches in depth at any point when tested as specified in 4.3.3.3.

3.6 Resistance to high temperature.-There shall be no blisters, cracks, or discoloration of the decorative laminate when the assembly is tested as specified in 4.3.4.

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3.7 Resistance to burning cigarettes (type III only).-The surface of type III assembly shall be completely resistant to the actual burning of a cigarette placed in contact with the surface and such exposure shall cause no cracks, crazing, discoloration, nor any delamination from the core, when tested as specified in 4.3.5.

3.8 Certification.-When specified, the contractor shall forward to the procuring agency with each shipment of table tops or furniture assemblies, including a specimen of the decorative laminate used, a certified statement (1) identifying the adhesive used in the core and in attaching the faces to the core, and (2) certifying that the table top or furniture assembly conforms to the requirements of this specification.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Sampling.-For Types I, II, and IV at least one assembly with an area of at least four square feet for each combination specified (the edging need not be applied to this sample), and a piece of decorative laminate with an area of at least two square feet for each combination specified, shall be used to determine conformance with this specification. For type III, a piece with an area of at least four square feet shall be submitted.

4.2 Inspection.-The samples shall be inspected for type of material, color, finish, defects, workmanship, and for conformity with all applicable requirements of this specification. Inspection shall also include packaging, packing, and marking for shipment.

4.3 Test procedures.

4.3.1 Conditioning.-The test specimens shall be conditioned and tested in accordance with the requirements of Federal Specification L-P-406, unless other conditions are specified in this specification or in the test method. In particular, specimens of thetable top or furniture assembly shall be kept at the same conditions at least 96 hours before test.

4.3.2 Impact resistance.-The specimen shall be a piece of the table top assembly, 6 by 12 inches. Lines, 2 inches apart, shall be drawn on the decorative laminate side of the specimen with a China marking pencil. The specimen shall be securely fastened to a level hardwood supporting surface. An 8-ounce (1.50 inches diameter) steel ball shall be dropped on the specimen onto the 10 points where the lines intersect. None of those points shall be less than 2.0 inches from the edge of the test specimen. The ball shall be dropped from a height of 24 inches (distance between the bottom of the ball and the decorative laminate). The decorative laminate shall be examined for cracks.

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4.3.3 Resistance to wetting and drying.

4.3.3.1 Type I.-Three specimens 1.5 inches square shall be given 5 cycles of the following exposure:

Period, hours	Temperature, °C.	Condition
4	97 - 100	Immersion in boiling water
20	60 ± 3	Less than 10 percent relative humidity in a circulating air oven.

4.3.3.2 Type II (5 ply plywood).-Three specimens 1.5 inches square shall be given 5 cycles of the following exposure:

Period, hours	Temperature, °C.	Condition
16	23 ± 1.1	Immersion in water
8	60 ± 3	Less than 10 percent relative humidity in a circulating air oven.

4.3.3.3 Type II (3 ply plywood), Type III, and Type IV.-Test specimen not less than 6 inch by 6 inch cut from separate panels shall be submerged in water at room temperature for a period of four (4) hours, removed and allowed to dry at a temperature between 70° to 100°F. for a period of 20 hours. The cycle shall then be repeated after which the panel is examined for delamination.

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4.3.4 Resistance to high temperature.-Two specimens 9 inches square shall be cut from the table top assembly. A flat-bottom round aluminum vessel having a capacity of approximately one pint, Fisher Scientific Company, Cat. No. 2-542 No-Bump Solution Evaporator, 600 milliliter, with the bottom sanded flat, or equivalent, shall be filled with wax or paraffin, heated to 180° ± 1°C. (356° ± 2°F.), and then placed on the center of the test specimen for 20 minutes. The surface of the specimen shall be inspected for blisters, cracks and discoloration.

4.3.5 Resistance to burning cigarettes.-Light a cigarette and place it laterally on the surface of a section of type III assembly and permit the lighted cigarette to burn completely. Examine the surface for cracks, crazing, discoloration and delamination.

5. PREPARATION FOR DELIVERY.

The definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 Level A.-To be inserted by the Department of Defense, if required.

5.1.2 Level C.-Unless otherwise specified in the contract or purchase order, the table top and furniture assemblies shall be packaged in accordance with accepted commercial practice.

5.2 Packing.

5.2.1 Level A.-To be inserted by the Department of Defense, if required.

5.2.2 Level B.-Unless otherwise specified, table top and furniture assemblies packaged as described in 5.1.3 shall be packed in domestic type wood-cleated fiberboard, wood-cleated-plywood, nailed wood, wirebound wood, or fiber boxes conforming to Federal Specifications PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-585, and LLL-B-636, respectively. The gross weight of wood or wood-cleated boxes shall not exceed approximately 100 pounds. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

5.2.3 Level C.-The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 Each table top and furniture assembly shall be marked on the back with a designation of type and the name of manufacturer.

5.3.2 In addition to the marking specified in 5.3.1 and any special marking required by the contract or order, marking shall be in accordance with Federal Standard No. 123.

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6. NOTES.

6.1 Intended use.-The table top and furniture assemblies covered by this specification are intended for use in living areas and cafeterias as table tops and counters, and in kitchens as working surfaces in the preparation of food. This specification does not cover table top assemblies for use in chemical laboratories. Type I table top assemblies should be used where exposure to water is frequent, as in installations near sinks.

6.2 Ordering data.-Purchasers should specify the type, size, color and pattern, finish and trim, and should exercise any desired options offered herein (see 1.2, 3.1, 3.2, 5.1, 5.2 and 5.3.2) and in the appplicable referenced documents covering the plywood and decorative laminate.

6.3 Transportation Description

6.3.1 The applicable transportation description, carload and truckload minimum weights to be added, is as follows:

Tops, table, wooden, covered with laminated
plastics or synthetic resin composition.
Carload weight 24,000 pounds, subject to
Rule 34, Uniform Freight Classification.

Tops, table, wooden, covered with laminated plastic.
Truckload weight 24,000 pounds.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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